

Universal Row Complexity of Simplicial Complexes

Common-Shell Threshold Systems and Bellman Universality

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Abstract

For a fixed ground set $[s]$, consider representations of labeled simplicial complexes of the form

$$K = \{J \subseteq [s] : A\mathbf{1}_J \leq b_K\},$$

where the nonnegative coefficient matrix A must be chosen once and for all, while only the right-hand side may depend on the target complex K . We determine the exact minimum number of rows required for such simultaneous universality. Writing

$$w_s = \binom{s}{\lfloor s/2 \rfloor},$$

we prove $u(s) = w_s$. In this sense the exact simultaneous fixed-bank row complexity is the Boolean-lattice width. The upper bound is realized by one positive integer matrix obtained from a symmetric-chain decomposition of the Boolean lattice, while the lower bound follows from shattering a maximum antichain by lower orthants.

We then impose the stronger requirement that all target-dependent quota vectors lie on one common ℓ_1 -shell. Remarkably, this costs no additional row:

$$u_\Sigma(s) = u(s) = w_s.$$

The extremal case requires a compression argument on a complete middle level that frees a balancing row without changing the represented complex.

Finally, the same universal coefficient system is lifted to one positive-cost periodic shortest-path system $\mathcal{U}_s$. For every labeled simplicial complex K on $[s]$, there is a state v_K such that the distinguished Bellman-optimal compatibility complex is exactly K , and all canonical witnesses lie at the same exact distance from the goal. The realization is extension-complete: after any prescribed valid lower truncation, every higher simplicial extension occurs in the same system. Thus one optimal fixed additive system simultaneously realizes the entire labeled simplicial universe, common-shell synchronization has zero row defect, and the resulting representation admits an exact Bellman-geometric realization.

1 Introduction

1.1 A simultaneous representation problem

Many representations of monotone or simplicial structures allow the coefficients themselves to depend on the target object. The present paper asks a different question. Fix s . Can one choose a single nonnegative matrix

$$A \in \mathbb{R}_{\geq 0}^{d \times s}$$

such that every labeled simplicial complex K on $[s]$ can later be programmed only by changing a right-hand side b_K ?

Equivalently, we ask for

$$\exists A_s \quad \forall K \in \text{Simp}([s]) \quad \exists b_K$$

such that

$$K = \{J \subseteq [s] : A_s \mathbf{1}_J \leq b_K\}.$$

The matrix is chosen before the target complex is known. We define the *simultaneous universal row complexity* $u(s)$ to be the minimum number of rows in such a matrix. The first main result determines this quantity exactly:

$$u(s) = \binom{s}{\lfloor s/2 \rfloor}.$$

Thus the width of the Boolean lattice is the exact cost of universal preprogramming in this additive lower-orthant model. Sperner's theorem supplies the classical width scale [1], while symmetric-chain decompositions provide the constructive decomposition used below [2].

1.2 Why this is not target-dependent threshold dimension

For a fixed target complex one may minimize the number of weighted or threshold components used to represent that particular object; such questions have the schematic form

$$\forall K \quad \exists A_K, b_K.$$

This belongs to the classical weighted-game and threshold-representation landscape [5]. In simple-game dimension theory, the weighted components are allowed to depend on the target game; the classical Taylor–Zwicker construction bounds their number by the number of maximal losing coalitions and hence, by Sperner's theorem, by the central-binomial scale w_s [5, 6]. Later work improves that target-dependent bound [6]. None of these formulations fixes one coefficient bank before the target is known.

Our problem reverses the first two quantifiers:

$$\exists A_s \quad \forall K \quad \exists b_K.$$

The same rows must support every labeled simplicial complex on $[s]$; only the quota vector is reprogrammed.

Fixed weights with variable thresholds are classical in threshold logic [4], and fixed matrices with varying right-hand sides are standard in parametric optimization [9]. The novelty here is the exact minimum row count under simultaneous fixed-bank universality over the full labeled simplicial target class.

1.3 Common-shell universality

We next require all quota vectors to have one common total mass,

$$\|b_K\|_1 = R \quad \text{for every } K.$$

Let $u_\Sigma(s)$ be the minimum row count under this restriction. A naive construction suggests that one additional balancing row may be necessary. The main theorem shows otherwise:

$$u_\Sigma(s) = u(s).$$

More precisely, one and the same optimal matrix realizes every labeled simplicial complex with

$$\|b_K\|_1 = w_s(2^s - 1).$$

Hence the common-shell row defect

$$\delta_\Sigma(s) := u_\Sigma(s) - u(s)$$

vanishes identically. The extremal case is not handled by ordinary padding; it requires a middle-level compression argument.

1.4 Bellman realization and extension completeness

The second part of the paper converts the coefficient system into one positive-cost periodic shortest-path system $\mathcal{U}_s$. For a target quota vector b , the canonical state is $v = -b$, and Bellman optimality reduces exactly to

$$A_s \mathbf{1}_J \leq b.$$

Therefore every labeled simplicial complex is realized literally as a distinguished compatibility complex in one fixed system for the given s , and all canonical witnesses lie at the same exact distance.

The universality is also conditional. After any valid lower truncation L through rank r has been fixed, every higher simplicial extension compatible with L appears in the same system. This yields complete extension fibers and, as a corollary, simultaneous all-rank saturation.

1.5 Research origin

The problem grew out of the author's broader study of deterministic pushdown normalization and shortest-completion geometry. More specifically, the route began with the author's completion-aware 2:1 and $p:q$ deterministic pushdown constructions, whose residual and shortest-completion viewpoints led to the rank-wise compatibility systems abstracted here. That program first produced arithmetic residual models and later periodic shortest-path systems exhibiting universal rank-wise compatibility fibers. The present paper asks whether dependence on a single rank can be removed and whether all simplicial orders can be realized simultaneously. The earlier work serves only as research provenance and motivation. The present paper is self-contained, and no unpublished result is used as a hypothesis.

1.6 Main contributions

The main contributions are:

1. exact simultaneous universal row complexity $u(s) = w_s$;
2. zero common-shell defect $u_\Sigma(s) = u(s)$;
3. exact fixed-system Bellman universality with one common canonical distance;
4. complete extension fibers for every lower truncation;
5. simultaneous all-rank saturation as a consequence of extension completeness.

Dedekind-capacity, topological universality, and uniform undecidability results derived later are consequences of this exact representation theorem rather than independent headline claims.

2 Definitions and Main Results

Let $[s] = \{1, \dots, s\}$, and let $\text{Simp}([s])$ denote the nonvoid downward-closed families of subsets of $[s]$. Ghost vertices are allowed unless stated otherwise. Throughout the paper, the rank of a face J means its cardinality $|J|$; thus a rank- k face has simplicial dimension $k - 1$.

For

$$A \in \mathbb{R}_{\geq 0}^{d \times s}, \quad b \in \mathbb{R}_{\geq 0}^d,$$

define

$$\mathcal{K}(A, b) = \{J \subseteq [s] : A \mathbf{1}_J \leq b\}.$$

Every such family is simplicial because $A \geq 0$.

Definition 2.1. The *simultaneous universal row complexity* is

$$u(s) = \min \left\{ d : \exists A \in \mathbb{R}_{\geq 0}^{d \times s} \forall K \in \text{Simp}([s]) \exists b_K \geq 0 : K = \mathcal{K}(A, b_K) \right\}.$$

The *common-shell universal row complexity* is

$$u_\Sigma(s) = \min \left\{ d : \exists A \in \mathbb{R}_{\geq 0}^{d \times s}, \exists R \geq 0 \forall K \in \text{Simp}([s]) \exists b_K \geq 0 : K = \mathcal{K}(A, b_K), \|b_K\|_1 = R \right\}.$$

Put

$$w_s = \binom{s}{\lfloor s/2 \rfloor}, \quad W_s = 2^s - 1, \quad R_s = w_s W_s.$$

Theorem 2.2 (Exact Universal Row Complexity and Zero Shell Defect). *For every $s \geq 2$,*

$$u(s) = u_\Sigma(s) = w_s.$$

Moreover, there exists

$$A_s \in \mathbb{Z}_{>0}^{w_s \times s}$$

such that for every $K \in \text{Simp}([s])$ there is $b_K \in \mathbb{Z}_{\geq 0}^{w_s}$ with

$$K = \mathcal{K}(A_s, b_K), \quad \|b_K\|_1 = R_s.$$

No nonnegative real coefficient matrix with fewer than w_s rows is universal. Consequently $\delta_\Sigma(s) = 0$.

This result is proved as theorem 4.5.

The notation $H(v, G)|_{I_\star}$, defined formally in Section 5, denotes the family of distinguished action subsets satisfying the corresponding Bellman optimality equality at v .

Theorem 2.3 (Universal Bellman Compatibility). *For every $s \geq 2$, there exists one positive-cost translation-periodic shortest-path system $\mathcal{U}_s$ with distinguished action set $I_\star = \{a_1, \dots, a_s\}$ such that for every $K \in \text{Simp}([s])$ there exists a state v_K satisfying*

$$H(v_K, G)|_{I_\star} = K.$$

The witness may be chosen so that

$$d((v_K, \mathbf{P}), G) = 1 + R_s$$

for every K .

This result is proved as theorem 5.5.

Theorem 2.4 (Universal Extension-Fiber Completeness). *Fix $s \geq 2$, $0 \leq r \leq s$, and a downward-closed lower truncation $L \subseteq 2_{\leq r}^{[s]}$. Let $P_{s,r}(L)$ be the poset of higher faces compatible with L . Then*

$$\text{ExtFib}_r(L) \cong \mathcal{J}(P_{s,r}(L)),$$

and the same fixed Bellman system realizes the entire fiber:

$$\text{ExtFib}_r^{\mathcal{U}_s}(L) = \text{ExtFib}_r(L).$$

Every member admits a witness at the same canonical distance $1 + R_s$.

This result is proved as theorem 6.2.

Corollary 2.5 (Simultaneous All-Rank Saturation). *For every $2 \leq k \leq s$, the same system $\mathcal{U}_s$ realizes every subset of the rank- k layer while the complete lower truncation through rank $k - 1$ is fixed. Hence*

$$|\text{RankFib}_k| = 2^{\binom{s}{k}}.$$

This result appears as corollary 6.3.

3 Universal Coefficient Construction

Fix $s \geq 2$. Choose a symmetric chain decomposition

$$2^{[s]} = C_1 \sqcup \dots \sqcup C_{w_s},$$

whose classical existence goes back to de Bruijn, van Ebbenhorst Tengbergen, and Kruyswijk [2]. For each chain C_q , choose an ordering $i_{q,1}, \dots, i_{q,s}$ of $[s]$ such that every member of C_q is an initial segment for the corresponding admissible cardinalities. Define the q th row of A_s by

$$a_{q,i_{q,j}} = 2^{s-j}.$$

Thus every row is a permutation of $2^{s-1}, \dots, 1$, and every row has total sum W_s .

3.1 Exact one-row separation

Let $E \in C_q$ with $|E| = m$, so

$$E = \{i_{q,1}, \dots, i_{q,m}\}.$$

Define

$$\tau_q(E) = W_s - 2^{s-m}.$$

Then $A_q \mathbf{1}_E = \tau_q(E) + 1$.

Lemma 3.1 (Exact chain separator). *For every $J \subseteq [s]$,*

$$A_q \mathbf{1}_J \leq \tau_q(E) \iff E \not\subseteq J.$$

Proof. If $E \subseteq J$, then $A_q \mathbf{1}_J \geq A_q \mathbf{1}_E = \tau_q(E) + 1$. Conversely, suppose $E \not\subseteq J$. At least one element of E is omitted. The smallest weight carried by an element of E is 2^{s-m} , whereas the total weight of all positions outside E is $2^{s-m} - 1$. Hence the largest score of a set omitting an element of E is at most

$$A_q \mathbf{1}_E - 2^{s-m} + (2^{s-m} - 1) = \tau_q(E).$$

□

3.2 Minimal nonfaces and canonical quotas

For $K \in \text{Simp}([s])$, let $\text{MinNF}(K)$ denote its set of minimal nonfaces. This is an antichain, so each chain C_q contains at most one minimal nonface. Define

$$b_{K,q}^0 = \begin{cases} \tau_q(E), & C_q \cap \text{MinNF}(K) = \{E\}, \\ W_s, & C_q \cap \text{MinNF}(K) = \emptyset. \end{cases}$$

Proposition 3.2 (Universal realization). *For every $K \in \text{Simp}([s])$,*

$$K = \{J \subseteq [s] : A_s \mathbf{1}_J \leq b_K^0\}.$$

Proof. If $J \in K$, no minimal nonface is contained in J , so every active row accepts J by lemma 3.1; inactive rows have quota W_s and are automatic. If $J \notin K$, then J contains some minimal nonface E , and the row corresponding to the chain of E rejects J by lemma 3.1. $\square$

Thus $u(s) \leq w_s$.

3.3 Canonical quota mass

The canonical quota total is

$$\|b_K^0\|_1 = w_s W_s - \sum_{E \in \text{MinNF}(K)} 2^{s-|E|}.$$

Define the canonical shell deficit by

$$\text{def}(K) := R_s - \|b_K^0\|_1.$$

Then

$$\text{def}(K) = \sum_{E \in \text{MinNF}(K)} 2^{s-|E|}.$$

If $|\text{MinNF}(K)| < w_s$, some row is inactive. Increasing its quota from W_s to $W_s + \text{def}(K)$ leaves the represented family unchanged and brings the total quota to R_s . Thus only the extremal case

$$|\text{MinNF}(K)| = w_s$$

remains.

3.4 Injectivity of subset scores

For every row q , the map $J \mapsto A_q \mathbf{1}_J$ is injective because the row entries are distinct powers of two. Hence

$$J \mapsto A_s \mathbf{1}_J$$

is injective on $2^{[s]}$.

4 Extremal Shell Compression and Optimality

4.1 Maximum antichains

Suppose $|\text{MinNF}(K)| = w_s$. Since $\text{MinNF}(K)$ is an antichain, equality holds in Sperner's theorem [1]. We use the LYM inequality in the form proved by Lubell [3].

Proposition 4.1 (Maximum antichains are middle levels). *If $\mathcal{A} \subseteq 2^{[s]}$ is an antichain with $|\mathcal{A}| = w_s$, then*

$$\mathcal{A} = \binom{[s]}{r}$$

for some middle rank r .

Proof. The LYM inequality gives

$$\sum_{E \in \mathcal{A}} \frac{1}{\binom{s}{|E|}} \leq 1.$$

Since $|\mathcal{A}| = w_s$ and $\binom{s}{|E|} \leq w_s$, equality must hold throughout, so every member lies on a maximum level. If s is even this level is unique.

Let $s = 2m + 1$. The two maximum levels are $\mathcal{L}_m$ and $\mathcal{L}_{m+1}$. Consider the bipartite inclusion graph between them; it is regular and connected. Put $X = \mathcal{A} \cap \mathcal{L}_m$. Since $\mathcal{A}$ is an antichain,

$$\mathcal{A} \cap \mathcal{L}_{m+1} \subseteq \mathcal{L}_{m+1} \setminus N(X).$$

For nonempty proper X , regularity gives $|N(X)| \geq |X|$. Equality would force $X \cup N(X)$ to be a union of connected components, contradicting connectedness. Hence $|N(X)| > |X|$, which forces $|\mathcal{A}| < w_s$. Therefore a maximum antichain cannot mix the two middle levels. $\square$

Consequently, in the extremal case

$$\text{MinNF}(K) = \binom{[s]}{r}, \quad K = 2_{\leq r-1}^{[s]}.$$

Every canonical row then has quota

$$\tau = W_s - 2^{s-r},$$

and

$$\text{def}(K) = w_s 2^{s-r}.$$

4.2 Two-for-one middle-level compression

Choose a chain C_{q_1} and write its rank- r member as

$$E = \{i_{q_1,1}, \dots, i_{q_1,r}\}.$$

Let

$$H = \{i_{q_1,1}, \dots, i_{q_1,r-1}\}, \quad \beta = A_{q_1} \mathbf{1}_H = 2^s - 2^{s-r+1}.$$

Lemma 4.2 (Compressed row). *Every J with $|J| \leq r - 1$ satisfies $A_{q_1} \mathbf{1}_J \leq \beta$, while every r -set F containing H violates this inequality.*

Proof. H contains the $r - 1$ largest row weights, so it maximizes the score among subsets of cardinality at most $r - 1$. If $F = H \cup \{x\}$, then positivity gives $A_{q_1} \mathbf{1}_F = \beta + a_{q_1,x} > \beta$. $\square$

There are $s - r + 1 \geq 2$ distinct r -sets containing H . Choose one $F \neq E$, and let q_2 be its chain. Modify the quotas by

$$\tilde{b}_{K,q_1} = \beta, \quad \tilde{b}_{K,q_2} = W_s,$$

and keep all other rows at the canonical value τ .

Proposition 4.3 (Extremal compression preserves the complex).

$$K = \{J \subseteq [s] : A_s \mathbf{1}_J \leq \tilde{b}_K\}.$$

Proof. If $|J| \leq r - 1$, the compressed row accepts by lemma 4.2, and every other active row is a canonical separator of an r -set, so all such J are accepted.

Now let $|J| \geq r$, and choose an r -subset $G \subseteq J$. If $G \notin \{E, F\}$, the canonical separator row belonging to the chain of G is unchanged and rejects J . If $G = E$ or $G = F$, then $H \subseteq G \subseteq J$, so the compressed row q_1 rejects J . Hence every set of cardinality at least r is rejected. $\square$

The compressed row changes from τ to β , so

$$\beta - \tau = 1 - 2^{s-r}.$$

Making row q_2 inactive adds 2^{s-r} , for a net gain of 1. Therefore

$$\|\tilde{b}_K\|_1 = w_s W_s + 1 - w_s 2^{s-r}.$$

The remaining deficit is

$$\Delta = w_s 2^{s-r} - 1 \geq 0.$$

Since row q_2 is inactive, increasing its quota by Δ preserves the represented family and yields total quota R_s . Therefore

$$u_\Sigma(s) \leq w_s.$$

4.3 Lower-orthant shattering

For $b \in \mathbb{R}^d$, write

$$O_b = \{x \in \mathbb{R}^d : x \leq b\}.$$

Lemma 4.4 (Lower-orthant shattering bound). *If $x_1, \dots, x_n \in \mathbb{R}^d$ are shattered by lower orthants, then $n \leq d$.*

Proof. For each i , shattering provides a lower orthant containing every point except x_i . Hence in some coordinate c_i , the point x_i is the unique strict maximum among all x_j . Two distinct points cannot be the unique strict maximum in the same coordinate, so $i \mapsto c_i$ is injective and $n \leq d$. $\square$

Let $A \in \mathbb{R}_{\geq 0}^{d \times s}$ be any simultaneous universal matrix. Choose a maximum antichain $\mathcal{A}$ with $|\mathcal{A}| = w_s$, and define score points

$$x_E = A \mathbf{1}_E, \quad E \in \mathcal{A}.$$

For any $S \subseteq \mathcal{A}$, set $K_S = \downarrow S$ if $S \neq \emptyset$, and $K_\emptyset = \{\emptyset\}$. Since $\mathcal{A}$ is an antichain,

$$K_S \cap \mathcal{A} = S.$$

Universality of A gives a lower orthant cutting out exactly S from the score points. Thus the score points are shattered. They are also pairwise distinct: if $x_E = x_F$ for distinct $E, F \in \mathcal{A}$, no lower orthant could contain one without the other. By lemma 4.4, $d \geq w_s$. Since every common-shell universal matrix is, after forgetting the shell constraint, an ordinary universal matrix, we also have

$$u_\Sigma(s) \geq u(s).$$

Combining these lower bounds with the constructions above gives equality in both models.

Theorem 4.5 (Exact Universal Row Complexity and Zero Shell Defect). *For every $s \geq 2$,*

$$u(s) = u_\Sigma(s) = w_s = \binom{s}{\lfloor s/2 \rfloor}.$$

Moreover, one positive integer matrix $A_s \in \mathbb{Z}_{>0}^{w_s \times s}$ simultaneously realizes every labeled simplicial complex on $[s]$, and every target admits a quota vector with

$$\|b_K\|_1 = w_s(2^s - 1).$$

Thus $\delta_\Sigma(s) = 0$. The row count is exact even against arbitrary nonnegative real coefficient matrices.

4.4 Full-vertex and one-vertex boundary cases

If ghost vertices are forbidden, then for $s \geq 3$ the same maximum-antichain shattering argument works using a middle level of rank at least two, so

$$u^{\text{fv}}(s) = u_\Sigma^{\text{fv}}(s) = w_s.$$

For $s = 2$,

$$u^{\text{fv}}(2) = 1, \quad u_\Sigma^{\text{fv}}(2) = 2.$$

Indeed, the only full-vertex complexes are $\{\emptyset, \{1\}, \{2\}\}$ and $2^{[2]}$. A single row $(1, 1)$ with thresholds 1 and 2 realizes the two targets. A one-row common-shell realization is impossible because equal ℓ_1 -mass would force the scalar thresholds to coincide. Two identical rows $(1, 1)$ with quota vectors $(1, 3)$ and $(2, 2)$ realize the two targets at common total 4.

For completeness, if $s = 1$ and ghosts are allowed,

$$u(1) = 1, \quad u_\Sigma(1) = 2.$$

A common-shell realization is given by

$$A = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad b_{\{\emptyset\}} = (0, 2)^T, \quad b_{2^{[1]}} = (1, 1)^T.$$

One common-shell row is impossible because the two scalar quota values would have to be equal.

5 Periodic Shortest-Path Lift and Exact Bellman Universality

Write

$$A_s = [u_1 \ \cdots \ u_s], \quad u_i \in \mathbb{Z}_{>0}^{w_s}.$$

Define the state space

$$V_s = \mathbb{Z}^{w_s} \times \{\mathbf{P}, \mathbf{C}\},$$

with goal $G = (0, \mathbf{C})$. For each $i \in [s]$, introduce a distinguished action

$$a_i : (x, \mathbf{P}) \rightarrow (x + u_i, \mathbf{P}), \quad c(a_i) = \|u_i\|_1.$$

Introduce a commit action

$$\gamma : (x, \mathbf{P}) \rightarrow (x, \mathbf{C}), \quad c(\gamma) = 1,$$

and completion moves

$$(x, \mathbf{C}) \rightarrow (x \pm e_q, \mathbf{C}), \quad q = 1, \dots, w_s,$$

each of cost 1. There is no return from $\mathbf{C}$ to $\mathbf{P}$. Denote the system by $\mathcal{U}_s$.

Proposition 5.1 (Exact completion metric). *For every $x \in \mathbb{Z}^{w_s}$,*

$$d((x, \mathbf{C}), G) = \|x\|_1.$$

Proof. Coordinatewise cancellation gives a path of length $\|x\|_1$, and every unit move changes the ℓ_1 -distance to the origin by at most one. $\square$

Proposition 5.2 (Exact prefix metric). *For every $x \in \mathbb{Z}^{w_s}$,*

$$d((x, \mathbf{P}), G) = 1 + \|x\|_1.$$

Proof. Committing immediately gives the upper bound. For an arbitrary path, suppose distinguished actions are used with multiplicities $n_i \geq 0$ before commit, and set $U = \sum_i n_i u_i \geq 0$. Their total cost equals $\|U\|_1$. The remaining path costs at least $1 + \|x + U\|_1$, so the total is at least

$$\|U\|_1 + 1 + \|x + U\|_1 \geq 1 + \|x\|_1$$

by the triangle inequality. $\square$

Thus no one-use gadget is required: distinguished actions may repeat arbitrarily.

Let

$$I_\star = \{a_1, \dots, a_s\}.$$

For $J \subseteq [s]$, define

$$U_J = \sum_{i \in J} u_i = A_s \mathbf{1}_J, \quad c(J) = \sum_{i \in J} \|u_i\|_1 = \|U_J\|_1.$$

We declare J Bellman-optimal at $(v, \mathbf{P})$ when

$$c(J) + d((v + U_J, \mathbf{P}), G) = d((v, \mathbf{P}), G).$$

The family of all such J is denoted $H(v, G)|_{I_\star}$. By proposition 5.2, this is equivalent to

$$\|U_J\|_1 + \|v + U_J\|_1 = \|v\|_1.$$

Lemma 5.3 (Coordinate identity). *For $u, b \in \mathbb{R}_{\geq 0}^{w_s}$,*

$$\|u\|_1 + \|u - b\|_1 = \|b\|_1 \iff u \leq b.$$

Proof. Coordinatewise, for $u_q, b_q \geq 0$,

$$u_q + |u_q - b_q| = b_q$$

if and only if $u_q \leq b_q$. Summing gives the claim. $\square$

Proposition 5.4 (Exact Bellman encoding). *For every $b \in \mathbb{R}_{\geq 0}^{w_s}$,*

$$H(-b, G)|_{I_\star} = \{J \subseteq [s] : A_s \mathbf{1}_J \leq b\}.$$

Proof. At $v = -b$, the Bellman equality becomes

$$\|U_J\|_1 + \|U_J - b\|_1 = \|b\|_1,$$

which is equivalent to $U_J \leq b$ by lemma 5.3. $\square$

If J is Bellman-optimal at $-b$, then every $I \subseteq J$ is Bellman-optimal because $U_I \leq U_J \leq b$. Therefore every ordering of the actions in J appears as an initial segment of a shortest path.

Theorem 5.5 (Universal Optimal-Compatibility Theorem). *For every $s \geq 2$, the fixed system $\mathcal{U}_s$ realizes every labeled simplicial complex on $[s]$: for every $K \in \text{Simp}([s])$ there exists $v_K \in \mathbb{Z}^{w_s}$ with*

$$H(v_K, G)|_{I_*} = K.$$

The canonical choice is $v_K = -b_K$, where b_K is the common-shell quota vector of theorem 4.5. Every canonical witness satisfies

$$d((v_K, P), G) = 1 + w_s(2^s - 1).$$

Proof. By theorem 4.5, choose b_K with $K = \{J : A_s \mathbf{1}_J \leq b_K\}$ and $\|b_K\|_1 = R_s$. Then proposition 5.4 gives the exact equality of complexes, and proposition 5.2 gives the common distance. $\square$

The complete equivalence chain is

$$J \in K \iff A_s \mathbf{1}_J \leq b_K \iff \|U_J\|_1 + \|U_J - b_K\|_1 = \|b_K\|_1 \iff J \in H(v_K, G)|_{I_*}.$$

6 Extension Fibers and Simultaneous All-Rank Saturation

For $0 \leq r \leq s$, write

$$2_{\leq r}^{[s]} = \{J \subseteq [s] : |J| \leq r\}.$$

A lower truncation through rank r is a downward-closed family $L \subseteq 2_{\leq r}^{[s]}$ containing $\emptyset$.

Define the compatible higher-face poset

$$P_{s,r}(L) = \{E \subseteq [s] : |E| > r, \forall F \subseteq E (|F| \leq r \Rightarrow F \in L)\},$$

ordered by inclusion, and the extension fiber

$$\text{ExtFib}_r(L) = \{K \in \text{Simp}([s]) : K \cap 2_{\leq r}^{[s]} = L\}.$$

Theorem 6.1 (Extension-Ideal Classification). *For every lower truncation L ,*

$$\text{ExtFib}_r(L) \cong \mathcal{J}(P_{s,r}(L)).$$

The bijection is $K \mapsto K \cap P_{s,r}(L)$, with inverse $I \mapsto L \cup I$.

Proof. If $K \in \text{ExtFib}_r(L)$, then every higher face of K lies in $P_{s,r}(L)$, and the higher part is downward closed inside that poset. Conversely, if I is an order ideal of $P_{s,r}(L)$, then $L \cup I$ is simplicial: a subface of size at most r lies in L by compatibility, while a higher subface lies in $P_{s,r}(L)$ and hence in I by ideality. The two maps are inverse. $\square$

The fiber is therefore a finite distributive lattice with meet and join given by intersection and union.

Define the system extension fiber

$$\text{ExtFib}_r^{\mathcal{U}_s}(L) = \{H(v, G)|_{I_\star} : H(v, G)|_{I_\star} \cap 2_{\leq r}^{[s]} = L\}.$$

Theorem 6.2 (Universal Extension-Fiber Completeness). *For every $s \geq 2$, every $0 \leq r \leq s$, and every lower truncation L ,*

$$\text{ExtFib}_r^{\mathcal{U}_s}(L) = \text{ExtFib}_r(L) \cong \mathcal{J}(P_{s,r}(L)).$$

Every member admits a canonical witness at distance

$$D_s = 1 + w_s(2^s - 1).$$

Proof. Every system compatibility family is simplicial. Conversely, every $K \in \text{ExtFib}_r(L)$ is realized by theorem 5.5, with canonical witness at distance D_s . $\square$

Thus Bellman optimality imposes no additional higher-order restriction beyond simplicial downward closure in the constructed system.

For $2 \leq k \leq s$, let

$$L_{k-1} = 2_{\leq k-1}^{[s]},$$

and define

$$\text{RankFib}_k = \left\{ K \cap \binom{[s]}{k} : K \in \text{ExtFib}_{k-1}(L_{k-1}) \right\}.$$

Every $A \subseteq \binom{[s]}{k}$ is realized by

$$K_A = L_{k-1} \cup A,$$

with no faces above rank k .

Corollary 6.3 (Simultaneous All-Rank Saturation). *For every $2 \leq k \leq s$,*

$$\text{RankFib}_k = 2^{\binom{[s]}{k}}, \quad |\text{RankFib}_k| = 2^{\binom{s}{k}}.$$

The same system $\mathcal{U}_s$ works for every k .

If k_s is a middle rank, then

$$\log_2 |\text{RankFib}_{k_s}| = w_s.$$

7 Full Metric-Shell Capacity and Entropy

Let

$$R_s = w_s(2^s - 1), \quad D_s = 1 + R_s,$$

and define the full prefix-phase metric shell

$$\mathbb{S}_s = \{v \in \mathbb{Z}^{w_s} : d((v, \mathbf{P}), G) = D_s\} = \{v : \|v\|_1 = R_s\}.$$

Proposition 7.1 (Arbitrary-state classification). *If $v \leq 0$ coordinatewise, then*

$$H(v, G)|_{I_\star} = \{J \subseteq [s] : A_s \mathbf{1}_J \leq -v\}.$$

If v has at least one positive coordinate, then

$$H(v, G)|_{I_\star} = \{\emptyset\}.$$

In particular, every prefix-phase state produces a simplicial compatibility complex.

Proof. The Bellman equality is coordinatewise. If $v \leq 0$, it holds exactly when $U_J \leq -v$. If some coordinate of v is positive, then every nonempty U_J is strictly positive in that coordinate, so equality fails there; only $J = \emptyset$ remains. $\square$

Theorem 7.2 (Full-Shell Universality). *For every $s \geq 2$,*

$$\{H(v, G)|_{I_*} : v \in \mathbb{S}_s\} = \text{Simp}([s]).$$

Proof. By proposition 7.1, every shell state produces a simplicial complex. Conversely, every $K \in \text{Simp}([s])$ has a canonical witness $v_K \in \mathbb{S}_s$ by theorem 5.5. $\square$

Let $M(s)$ denote the Dedekind number, equivalently the number of antichains in $2^{[s]}$. Since we count nonvoid simplicial complexes, their number is $M(s) - 1$. Define the semantic shell capacity by

$$\text{Cap}(\mathbb{S}_s) = |\{H(v, G)|_{I_*} : v \in \mathbb{S}_s\}|.$$

Then

$$\text{Cap}(\mathbb{S}_s) = M(s) - 1.$$

This is semantic capacity, not geometric shell cardinality.

We use the phrase *semantic entropy* only as shorthand for the base-two logarithm of the number of distinct compatibility complexes; no probabilistic or Shannon-theoretic structure is intended. Kleitman's classical asymptotic solution of Dedekind's problem gives [10]

$$\log_2 M(s) = (1 + o(1))w_s.$$

Hence

$$\log_2 \text{Cap}(\mathbb{S}_s) = (1 + o(1))w_s$$

and therefore

$$u(s) \sim \log_2 \text{Cap}(\mathbb{S}_s).$$

The same middle-layer width controls both quantities, but no generic dimension-entropy law is claimed.

Moreover, for a middle rank k_s ,

$$\frac{\log_2 |\text{RankFib}_{k_s}|}{\log_2 \text{Cap}(\mathbb{S}_s)} \rightarrow 1.$$

Thus complete all-order universality strengthens exact structure without changing the leading first-order information scale already visible at the middle layer.

Let C_s be the number of labeled simplicial complexes on $[s]$ containing every singleton. Inclusion-exclusion over ghost vertices gives

$$C_s = \sum_{j=0}^s (-1)^j \binom{s}{j} (M(s-j) - 1).$$

The number with at least one ghost vertex is at most $s(M(s-1) - 1)$. Since $w_{s-1} = (1/2 + o(1))w_s$ and $\log_2 M(t) = (1 + o(1))w_t$, we have $M(s-1)/M(s) \rightarrow 0$, hence

$$C_s = (1 - o(1))M(s), \quad \log_2 C_s = (1 + o(1))w_s.$$

8 Topological Universality and Effective Consequences

Every finite labeled simplicial complex occurs literally as a distinguished compatibility complex. Consequently, across the uniform family $\{\mathcal{U}_s\}_{s \geq 2}$, every value of simplicial dimension, f -vector, Euler characteristic, homology or cohomology group, Betti-number pattern, torsion pattern, and finite simplicial homotopy type that is attained by a finite simplicial complex is also attained by a compatibility complex. In particular, every finitely presented group occurs as a fundamental group, as recorded below.

Corollary 8.1 (Distance does not determine topology). *Let $\mathcal{I}$ be any invariant nonconstant on $\text{Simp}([s])$. Then there exist $v_1, v_2 \in \mathbb{S}_s$ with equal distance D_s from the goal but*

$$\mathcal{I}(H(v_1, G)|_{I_*}) \neq \mathcal{I}(H(v_2, G)|_{I_*}).$$

Every finitely presented group occurs as the fundamental group of a finite two-dimensional simplicial complex; see, for example, standard presentation-complex treatments such as [11].

Corollary 8.2 (Fundamental-group universality). *For every finitely presented group Γ , there exist $s \geq 2$ and $v \in \mathbb{Z}^{w_s}$ such that*

$$\pi_1(|H(v, G)|_{I_*}) \cong \Gamma.$$

The realizing compatibility complex may be chosen two-dimensional, and v may be chosen as a canonical common-shell witness.

For algorithmic consequences, fix for each s a standard recursive symmetric-chain decomposition. Then $s \mapsto A_s$ is computable, and from a finite labeled simplicial complex K one can effectively enumerate minimal nonfaces, assign canonical quotas, perform ordinary shell padding or the explicit extremal compression, compute b_K , and output $v_K = -b_K$.

Proposition 8.3 (Effective universality). *There is an effective transformation*

$$K \longmapsto (s, v_K)$$

such that

$$H(v_K, G)|_{I_*} = K, \quad d((v_K, P), G) = D_s.$$

For fixed s , the semantic universe is finite, with $M(s) - 1$ objects, so no fixed- s undecidability claim is made. Undecidability can arise only when s varies over the uniform family.

Theorem 8.4 (Effective transfer principle). *Let $\mathcal{P}$ be a property of finite simplicial complexes. If deciding $\mathcal{P}$ for finite simplicial complexes is undecidable, then deciding whether $H(v, G)|_{I_*}$ has property $\mathcal{P}$ is undecidable across the uniform family $\{\mathcal{U}_s\}_{s \geq 2}$.*

Proof. Given K , compute (s, v_K) using proposition 8.3. A decision procedure for the Bellman version would decide $\mathcal{P}$ for K . $\square$

Triviality of the fundamental group is undecidable for finite simplicial complexes, and this remains true in dimension two [12, 13]. Hence simple connectivity is undecidable across the uniform Bellman family. The reduction already uses canonical witnesses on their prescribed common-distance shells.

These topological and algorithmic statements are consequences of exact effective universality, not the novelty center of the paper.

9 Relation to Previous Work and Novelty Boundary

The results combine several classical ingredients. The novelty claim is therefore attached to the complete simultaneous theorem rather than to these ingredients individually.

Fixed weights with variable thresholds are classical [4], and fixed coefficient matrices with varying right-hand sides are standard in parametric optimization [9]. Target-dependent weighted representations and dimensions belong to the classical simple-game framework [5]. In that theory, each target may choose its own weighted components; the Taylor–Zwicker construction gives a bound by the number of maximal losing coalitions and therefore a central-binomial upper bound, with subsequent improvements known [6]. Our exact value w_s should not be conflated with that classical bound: here w_s is the minimum size of a single bank of coefficient rows that is fixed before the target complex is known. We make no novelty claim for the target-dependent paradigms themselves.

Sperner’s theorem, the LYM inequality, and symmetric-chain decompositions are classical [1, 3, 2]. What is new in the present problem is that the Boolean-lattice width is exactly the minimum simultaneous universal row complexity: symmetric chains provide the construction, while a maximum antichain gives the lower-orthant obstruction.

Arbitrary finite simplicial complexes are already known to admit threshold-like or convex-geometric realizations in other models. Pakianathan and Winfree develop quota and threshold complexes [7], while Frick and Sanyal show that every simplicial complex can be realized as a Minkowski complex [8]. Accordingly, we do not claim novelty for the bare existence of a threshold-like realization of every finite simplicial complex.

The simultaneous restrictions imposed here are different: the ground set is fixed; one additive coefficient matrix is chosen for all targets; only the quota vector varies; the number of common rows is minimized; and all targets are synchronized to the same quota mass. The resulting zero shell defect

$$u_{\Sigma}(s) = u(s) = w_s$$

is a separate strengthening of ordinary universal representability.

Bellman’s optimality equation, translation-periodic shortest-path systems, and the ℓ_1 metric are standard. The contribution is their exact conjunction with the optimal simultaneous coefficient system: for every fixed s , one system $\mathcal{U}_s$ is chosen independently of the target, and

$$H(v_K, G)|_{I_*} = K$$

holds literally for every labeled simplicial complex K , with all canonical target states at one exact distance. The same system is extension-complete.

To the best of our knowledge, we are not aware of a previous construction establishing this complete conjunction: an exact minimum-row coefficient system fixed in advance for all labeled simplicial complexes on a given ground set, common-shell synchronization without increasing that minimum, and exact realization of the resulting complete family as Bellman-optimal compatibility complexes in one fixed positive-cost periodic shortest-path system, together with complete extension fibers. The priority claim of this paper is restricted to this combined theorem package.

The equality $u(s) = w_s$ is optimal only for the simultaneous nonnegative additive lower-orthant representation model defined here. No claim is made that the common radius R_s is minimum, that the coefficient magnitudes are optimal, or that w_s is the minimum dimension of an arbitrary Bellman architecture capable of realizing all complexes.

The discovery path began in the author’s earlier work on deterministic pushdown normalization, residual completion, and periodic shortest-path compatibility. Its role here is research

provenance only: no unpublished theorem is assumed, and no unpublished manuscript by the author is cited in the bibliography.

10 Conclusion

We introduced the simultaneous universal row-complexity problem for labeled simplicial complexes and determined its exact value. For every $s \geq 2$,

$$u(s) = \binom{s}{\lfloor s/2 \rfloor}.$$

The upper bound is realized by an explicit positive integer matrix built from a symmetric-chain decomposition of the Boolean lattice, and the matching lower bound follows from maximum-antichain trace shattering by lower orthants.

The stronger common-shell requirement costs no additional row:

$$u_{\Sigma}(s) = u(s).$$

Every target admits a quota vector of total mass

$$\binom{s}{\lfloor s/2 \rfloor} (2^s - 1).$$

The extremal middle-level compression proves the zero-defect identity $\delta_{\Sigma}(s) = 0$.

The same optimal coefficient system lifts to a fixed positive-cost translation-periodic shortest-path system $\mathcal{U}_s$ in which every labeled simplicial complex occurs exactly as a distinguished Bellman compatibility complex. All canonical witnesses lie at the common distance

$$1 + \binom{s}{\lfloor s/2 \rfloor} (2^s - 1).$$

The realization is extension-complete, and simultaneous all-rank saturation follows.

At the level of the full common-distance shell, the semantic image is exactly $\text{Simp}([s])$, so the exact semantic shell capacity is $M(s) - 1$. The classical Dedekind asymptotic gives

$$\log_2(M(s) - 1) \sim \binom{s}{\lfloor s/2 \rfloor}.$$

Thus the same Boolean-lattice width governs exact simultaneous universal row complexity, exact complete freedom of a middle rank, and the first-order logarithmic size of the full simplicial universe.

Finally, because every finite simplicial complex occurs literally and effectively, all finite simplicial homotopy types and finitely presented fundamental groups occur across the uniform family, and classical undecidable properties transfer when the ground-set size varies. These are downstream consequences of the main representation theorem.

The central conclusion is

$$u_{\Sigma}(s) = u(s) = \binom{s}{\lfloor s/2 \rfloor}.$$

A single optimal additive coefficient system can therefore be fixed in advance for the entire labeled simplicial universe, equal-total synchronization introduces no additional row complexity,

and the resulting complete family admits an exact fixed-system realization in Bellman optimality geometry.

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